JULIA PINEDO

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EDUCATION

MONTERREY INSTITUTE OF TECHNOLOGY AND HIGHER EDUCATION Remote
Master's Degree in Applied Artificial Intelligence Sep 2023 - Dec 2025
Final Grade: 98/100
Chief of Staff, Student Council (2025): Elected representative for the Applied AI student cohort.
Relevant Coursework: Data Science; Artificial Intelligence; Reinforcement Learning; Software Testing; Software Design.

TSINGHUA UNIVERSITY Remote
Academic Exchange Program Feb 2025 - Jun 2025
Coursework: Reinforcement Learning
- Completed a course on sequential decision-making, policy iteration, and value-based methods.
- Developed a final project to optimize traffic light control at high-congestion intersections using SUMO and Python.

ARIZONA STATE UNIVERSITY Remote
Academic Exchange Program Apr 2024 - Jul 2024
Coursework: Advanced Innovation
- Completed an intensive course on innovation ecosystems, design thinking, and strategic foresight.
- Collaborated with international peers to apply innovation frameworks to global challenges.

AUTONOMOUS UNIVERSITY OF BAJA CALIFORNIA Tecate, BC
Bachelor of Science in Mechatronics Engineering Aug 2018 - Dec 2022
Final Grade: 93/100
Merit Scholarship Recipient in 2022 (ranking in the top 1% of the student body).
Relevant Coursework: Linear Algebra; Numerical Methods; Object-Oriented Programming; Visual Programming.

WORK EXPERIENCE

INSULET CORPORATION Tijuana, BC
Senior Software Test Engineer Oct 2026 – Present
- Leading automation testing initiatives across multiple product features, expanding automated test coverage for complex scenarios.
- Developing AI-driven testing strategies to enable smarter test automation, including computer vision-based testing scenarios.
- Collaborating with software developers, systems engineers, and DQA teams to strengthen quality throughout the development lifecycle.

Software Test Engineer Mar 2025 – Oct 2026
- Designed and maintained end-to-end automated test suites using Python to enhance test coverage and reduce manual testing time.
- Actively contributed to AI and innovation initiatives by researching and evaluating computer vision and OCR-based solutions to enhance automated validation, identify testing gaps, and improve test coverage and efficiency.
- Partnered with developers and product owners to ensure alignment with business goals and compliance requirements.
- Documented complex testing scenarios and results with precision, improving cross-team communication and onboarding of new QA team members.

Associate Test Engineer Aug 2023 – Mar 2025
- Executed comprehensive manual and automated test cases, contributing to the successful delivery of application updates.
- Developed, refactored, and participated in code reviews for Python-based test scripts, increasing automation reliability.
- Maintained and supported CI/CD workflows by assisting with test integration and debugging, contributing to a 20% improvement in test run reliability.
- Documented test results, procedures, and known issues clearly and concisely, improving the stability of the deployment pipeline through better CI/CD test integration.

Software Tester Trainee (part-time) Mar 2023 – Aug 2023

- Developed foundational software testing skills through hands-on experience in defect tracking, test documentation, and exploratory testing.
- Authored basic test documentation and manual test cases, increasing test traceability and team knowledge sharing.
- Built Python scripts for automation under mentorship, gaining exposure to CI/CD workflows and automation best practices.
- Contributed to an improved testing process by helping the team maintain consistent test documentation.

RESEARCH EXPERIENCE

UNIVERSITY OF CALIFORNIA, SAN DIEGO La Jolla, CA

Undergraduate Research Assistant Aug 2022 - May 2023

- Contributed to the development of a Python-based optimization model for power systems using SWITCH, focusing on the integration of renewable energy and green hydrogen.
- Applied data science techniques to improve power grid simulations, identifying emissions reduction opportunities through scenario analysis.
- Conducted a comprehensive literature review on sustainable grid technologies and structured research questions to guide model design and validation.
- Supported strategic planning for renewable energy adoption in power grids.

MONTERREY INSTITUTE OF TECHNOLOGY AND HIGHER EDUCATION Remote

Research Intern in "Programa Delfín" June 2022 - Aug 2022

- Developed and implemented a Python-based genetic algorithm to optimize the structure of an artificial neural network for predictive modeling.
- Leveraged Jupyter Notebook for experimentation and Visual Studio for debugging, achieving improvements in training convergence and model accuracy.
- Presented research at an international conference, showcasing the practical application of evolutionary computing in machine learning.
- Achieved a measurable improvement in predictive performance by applying evolutionary strategies to neural network optimization.

UNIVERSITY OF CALIFORNIA, SAN DIEGO Remote

Research Intern in "ENLACE Summer Research Program" June 2021 - Aug 2021

- Investigated the use of machine learning models (linear regression, SVR, and MLP) to predict the physical properties of Heusler compounds.
- Conducted model evaluation and cross-validation using Python to assess the prediction of formation energy and quantum spin characteristics.
- Delivered final research findings at a symposium, contributing to early-stage efforts in AI-assisted materials discovery.
- Built foundational predictive models to accelerate the screening process for quantum materials with specific electronic properties.

PROFESSIONAL AFFILIATIONS

LATIN AMERICAN SOCIETY OF SCIENCE AND TECHNOLOGY (SOLACyT) Remote

Subcoordinator of the Design Committee from the Network of Youth Scientific Disseminators Apr 2023 - Present

- Founding member of the Network of Youth Scientific Disseminators.
- Managed project timelines and coordinated with other sub-committees, ensuring the on-time delivery of visual assets used in national-level workshops, social media campaigns, and dissemination events.
- Played a key role in scaling the network's visual identity and engagement efforts, strengthening its mission to promote science and technology education in underserved communities.

CERTIFICATES

Scrum Master Professional Certificate Issued in 2025

Online course by Certiprof End date: Apr 2028

High Specialty Certificate: Data Visualization Issued in 2025
Online course by Monterrey Institute of Technology and Higher Education End date: Apr 2030

High Specialty Certificate: Internet of Things and Sensor Networks Issued in 2024
Online course by Monterrey Institute of Technology and Higher Education End date: Dec 2027

Introduction to Programming in Python I: Learning to Program with Python Issued in 2024
Online course by Pontifical Catholic University of Chile (No end date)

High Specialty Certificate: Artificial Intelligence and Machine Learning Issued in 2024
Online course by Monterrey Institute of Technology and Higher Education End date: Aug 2027

High Specialty Certificate: Data Science Issued in 2024
Online course by Monterrey Institute of Technology and Higher Education End date: Jan 2027

Data Analysis Using Python Issued in 2023
Online course by IBM (No end date)

Getting Started with Enterprise Data Science Issued in 2023
Online course by IBM (No end date)

Python 101 for Data Science Issued in 2023
Online course by IBM (No end date)

Full Stack Engineer Bootcamp Issued in 2023
Online course by ITJ and UC San Diego Extended Studies (No end date)

Learning How to Learn: Powerful mental tools to help you master tough subjects Issued in 2020
Online course by Coursera (No end date)

HONORS AND AWARDS

1st PLACE WINNER AT THE 9th STATE MEETING OF YOUNG RESEARCHERS Mexicali, BC
Autonomous University of Baja California Nov 2022

- Winner in the CONAHCYT "Engineering and Industry" undergraduate category at the esteemed 9th State Meeting of Young Researchers in Baja California, as a result of presenting the project titled "Tuning parameters of a Neural Network through a Genetic Algorithm", advised by Dr. Arturo Sinué Ontiveros Zepeda.

ADDITIONAL

Technical Skills: Advanced in Python, Jupyter Notebook, Visual Studio Code, Git, Manual Testing, Test Automation, CI/CD Pipeline; Proficient in Postman, FastAPI, Machine Learning, Reinforcement Learning, Data Analysis, REST APIs.
Soft Skills: Leadership; Team coordination; Communication; Creativity; Visual thinking; Time management; Critical thinking; Attention to detail; Collaboration in Cross-functional teams.
Languages: Full professional proficiency in English; Native proficiency in Spanish; Elementary proficiency in German; Elementary proficiency in Chinese.